Arduino WorkShop

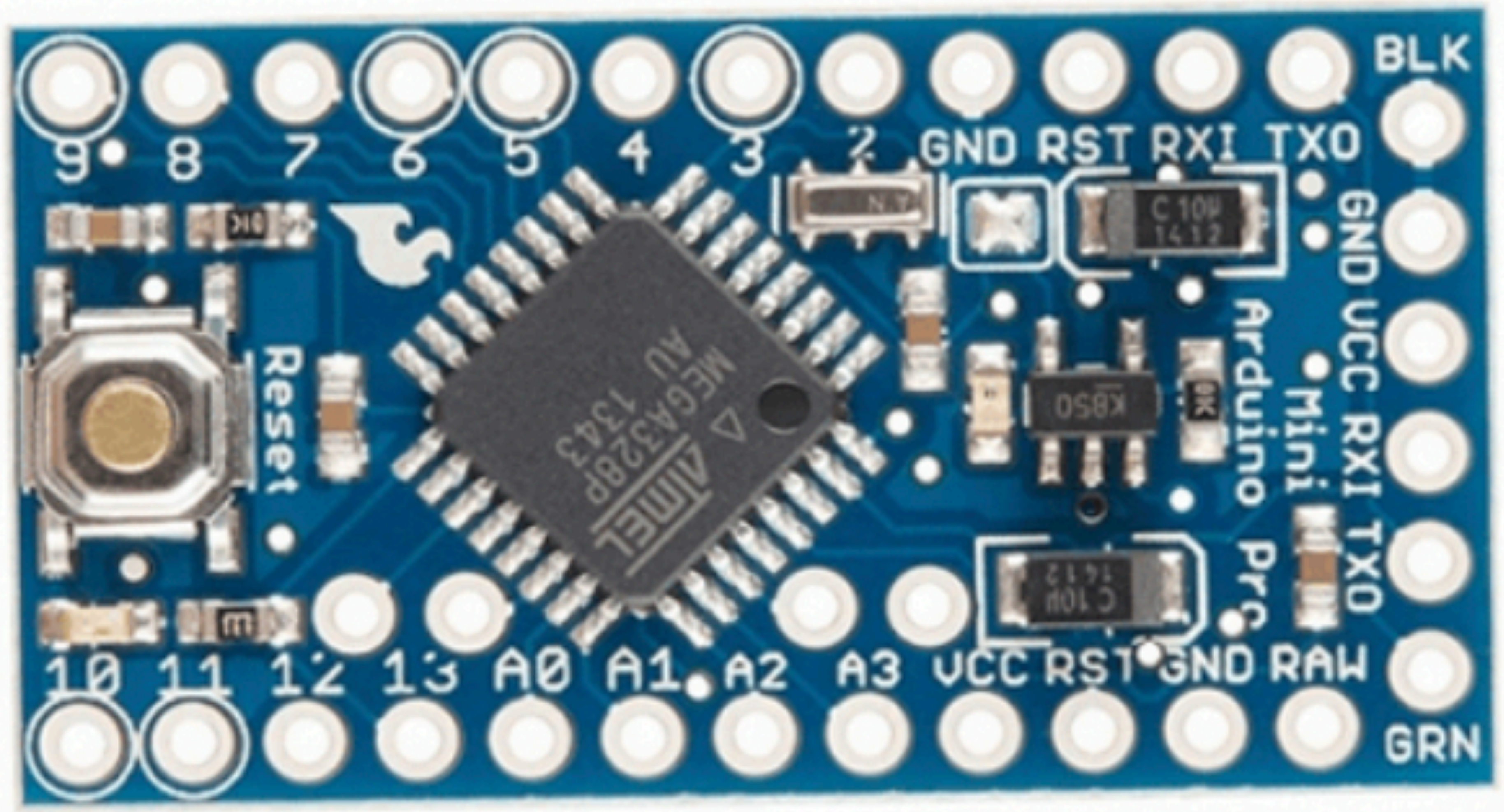
What Is Arduino ??

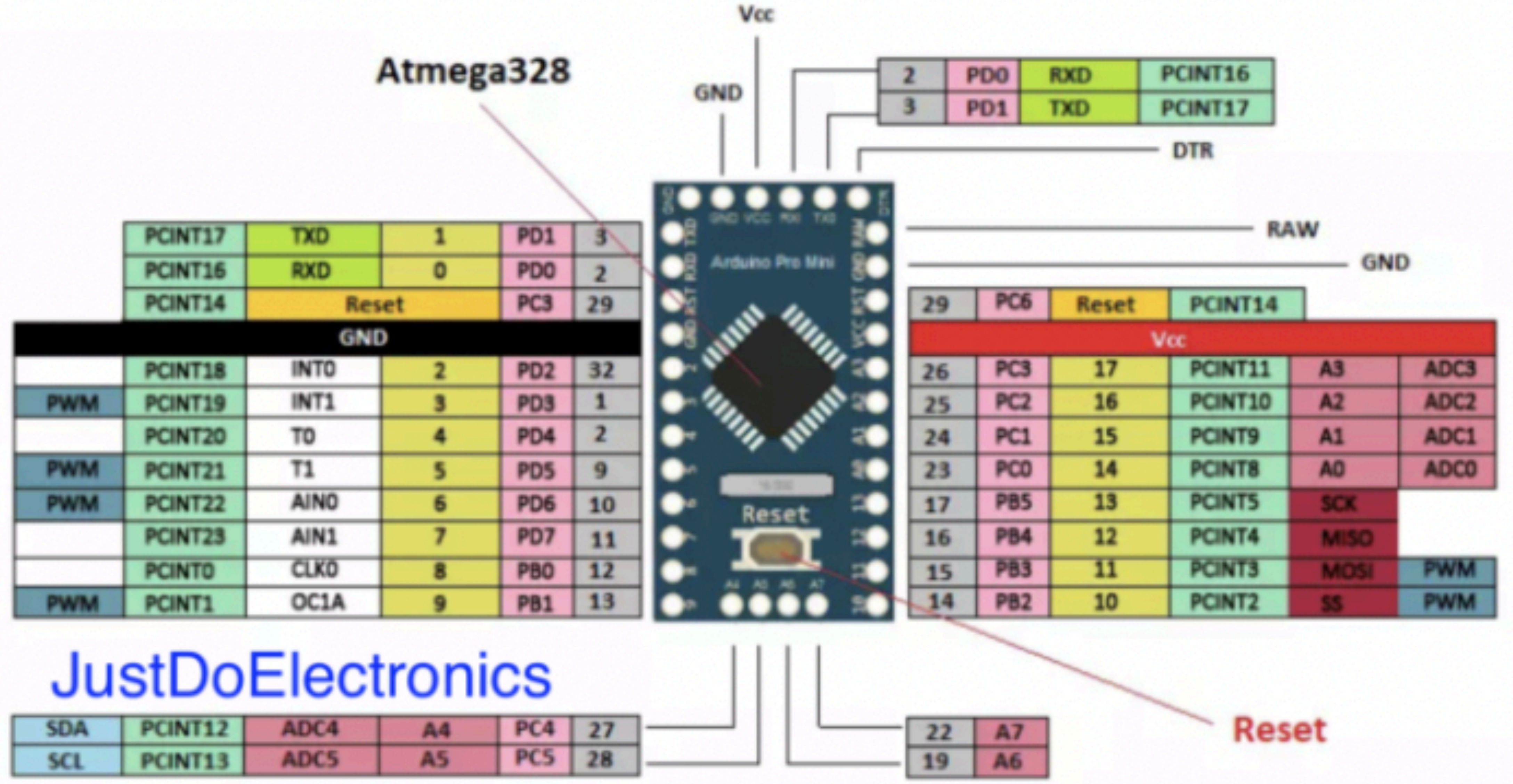
The Arduino platform has become quite popular with people just starting out with electronics, and for good reason. Unlike most previous programmable circuit boards, the Arduino does not need a separate piece of hardware (called a programmer) in order to load new code onto the board --

you can simply use a USB cable. Additionally, the Arduino IDE uses a simplified version of C++, making it easier to learn to program. Finally, Arduino provides a standard form factor that breaks out the functions of the micro-controller into a more accessible package.

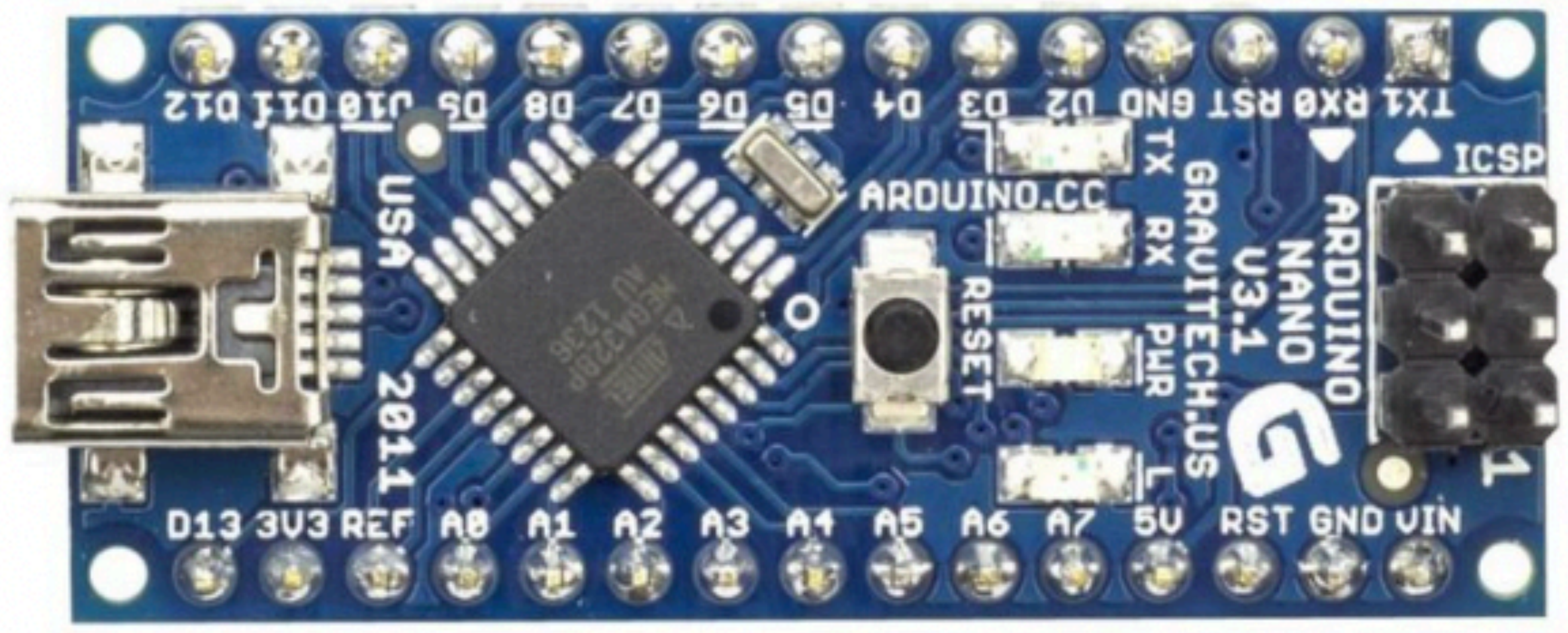
Arduino Type

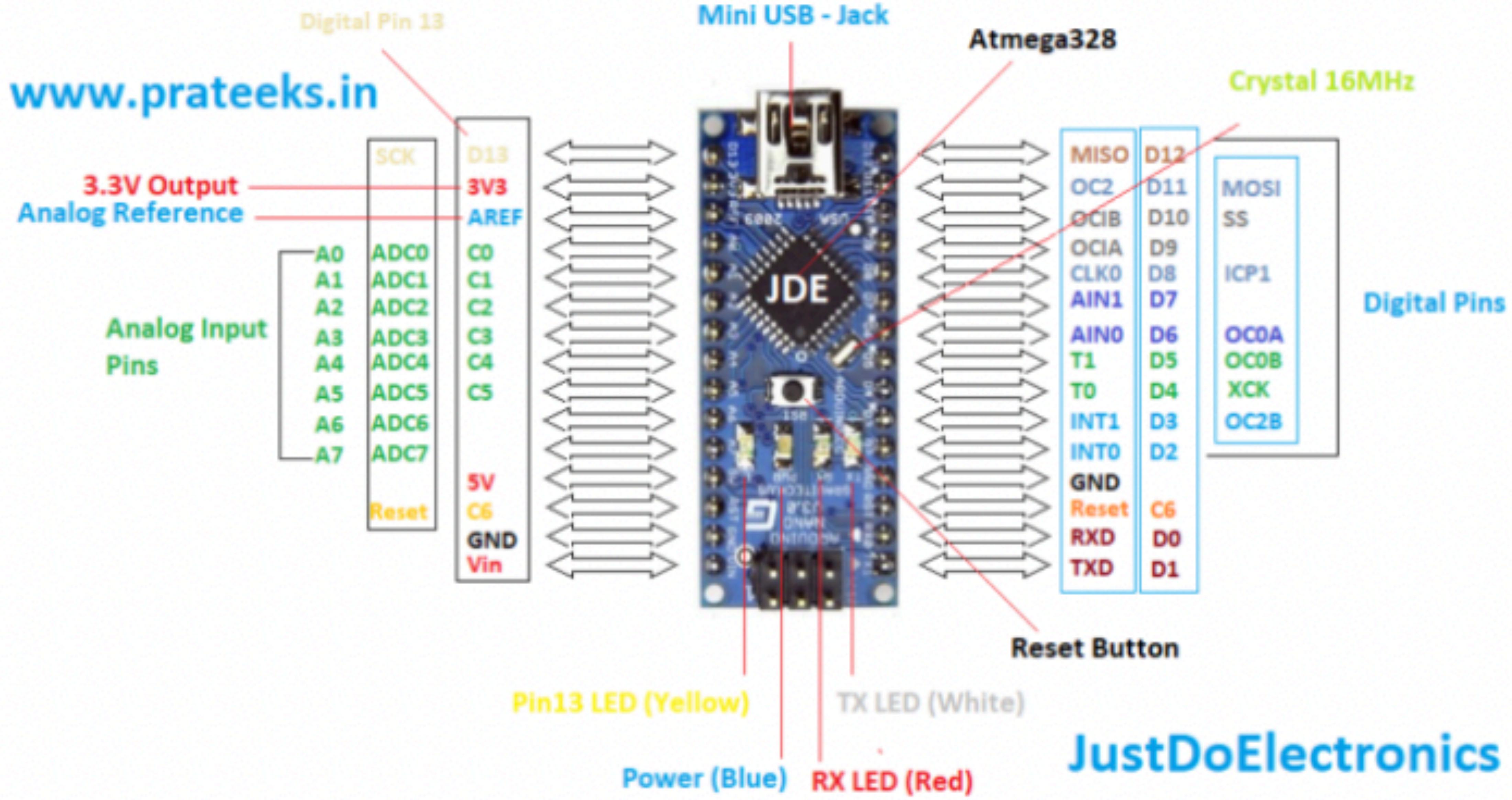
Arduino Pro Min



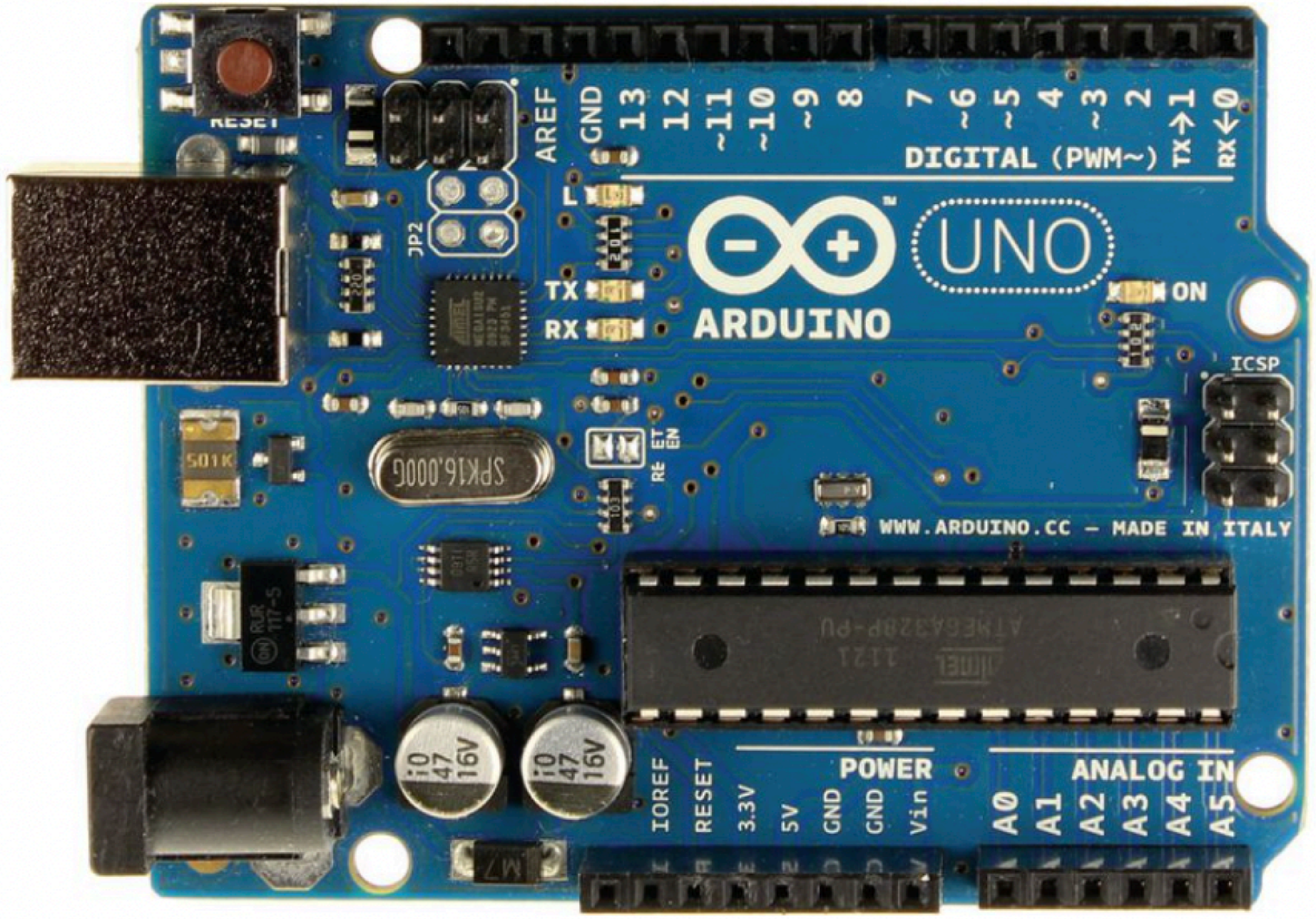


Arduino Nano

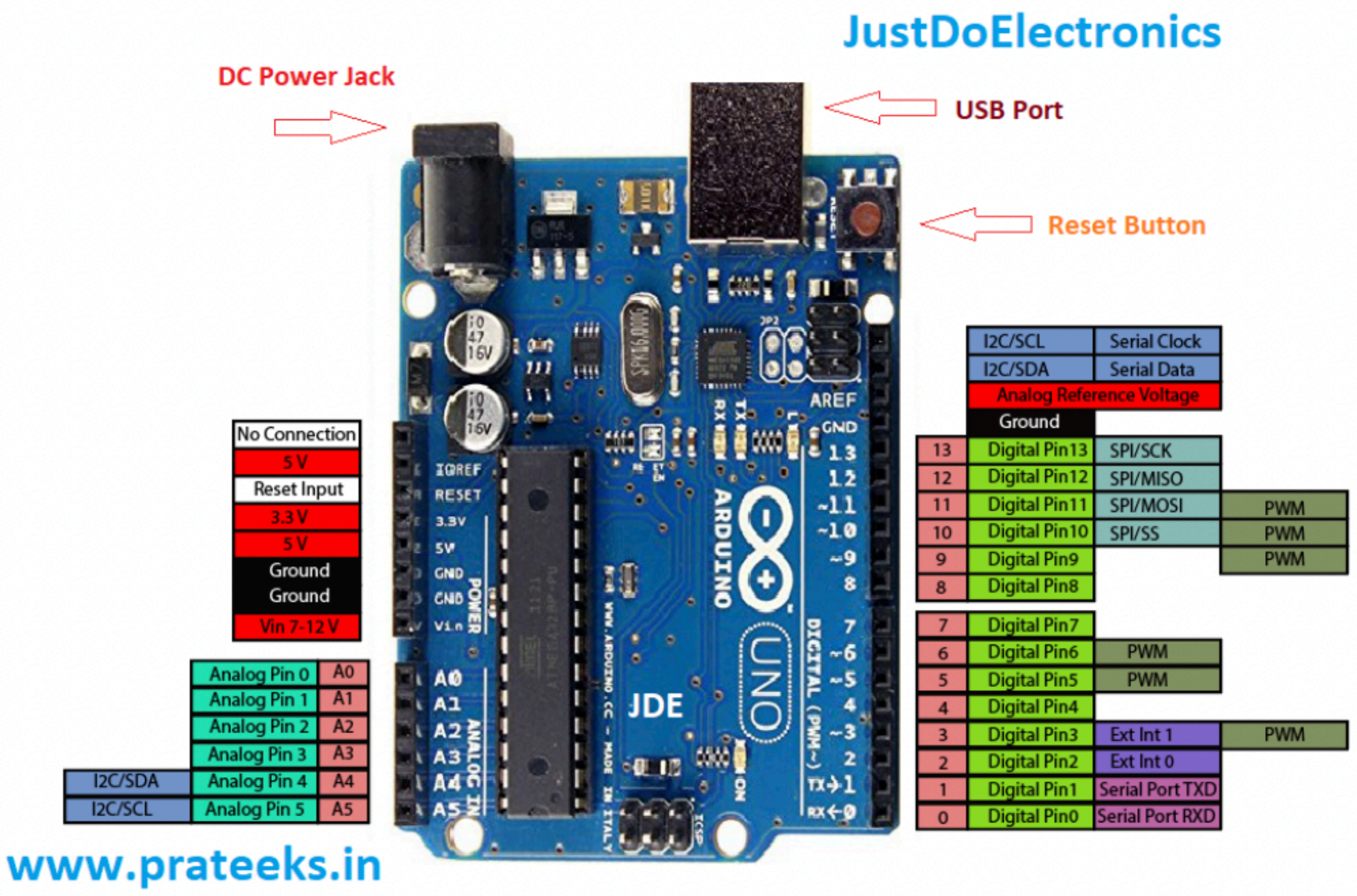




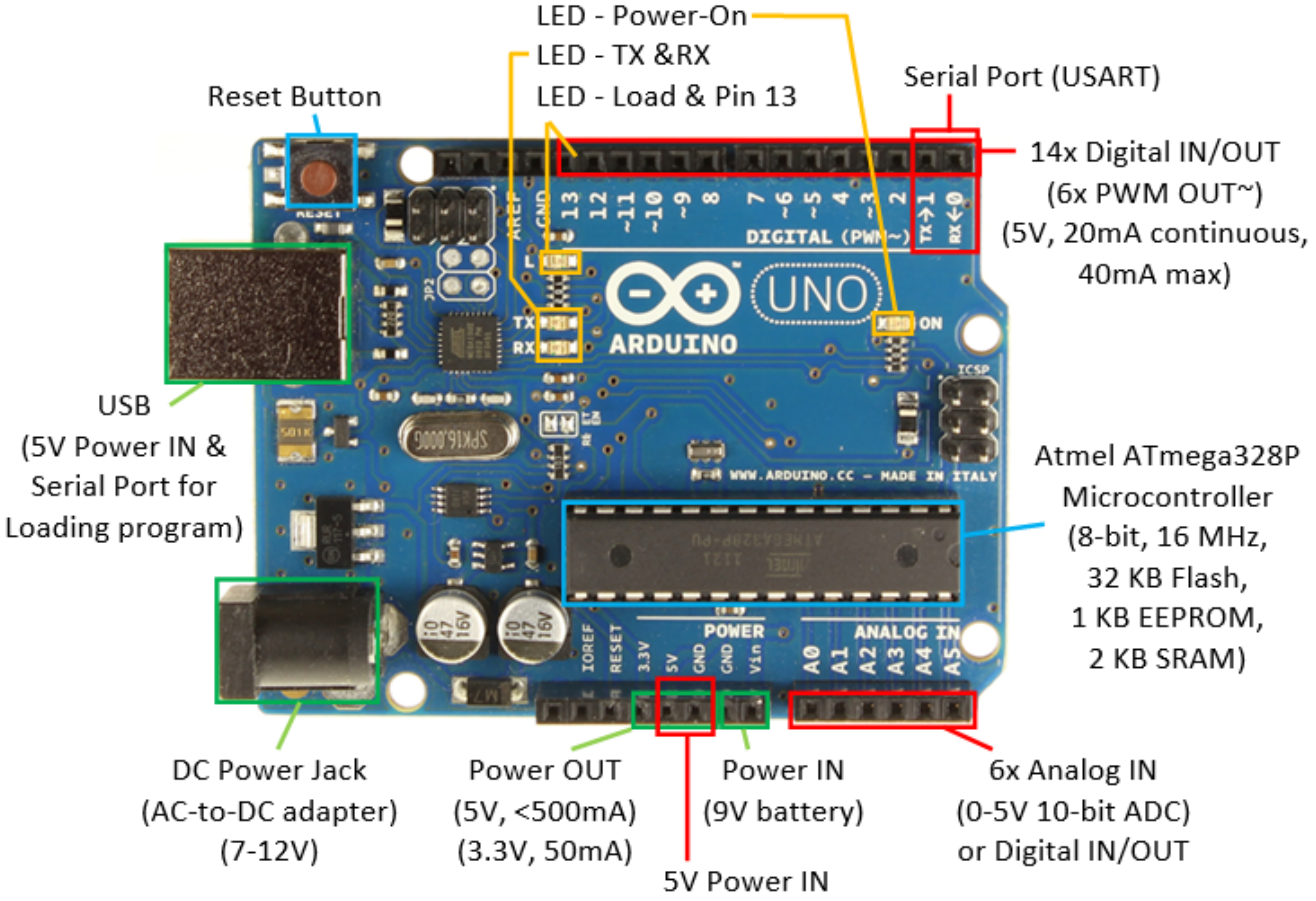
Arduino UNO



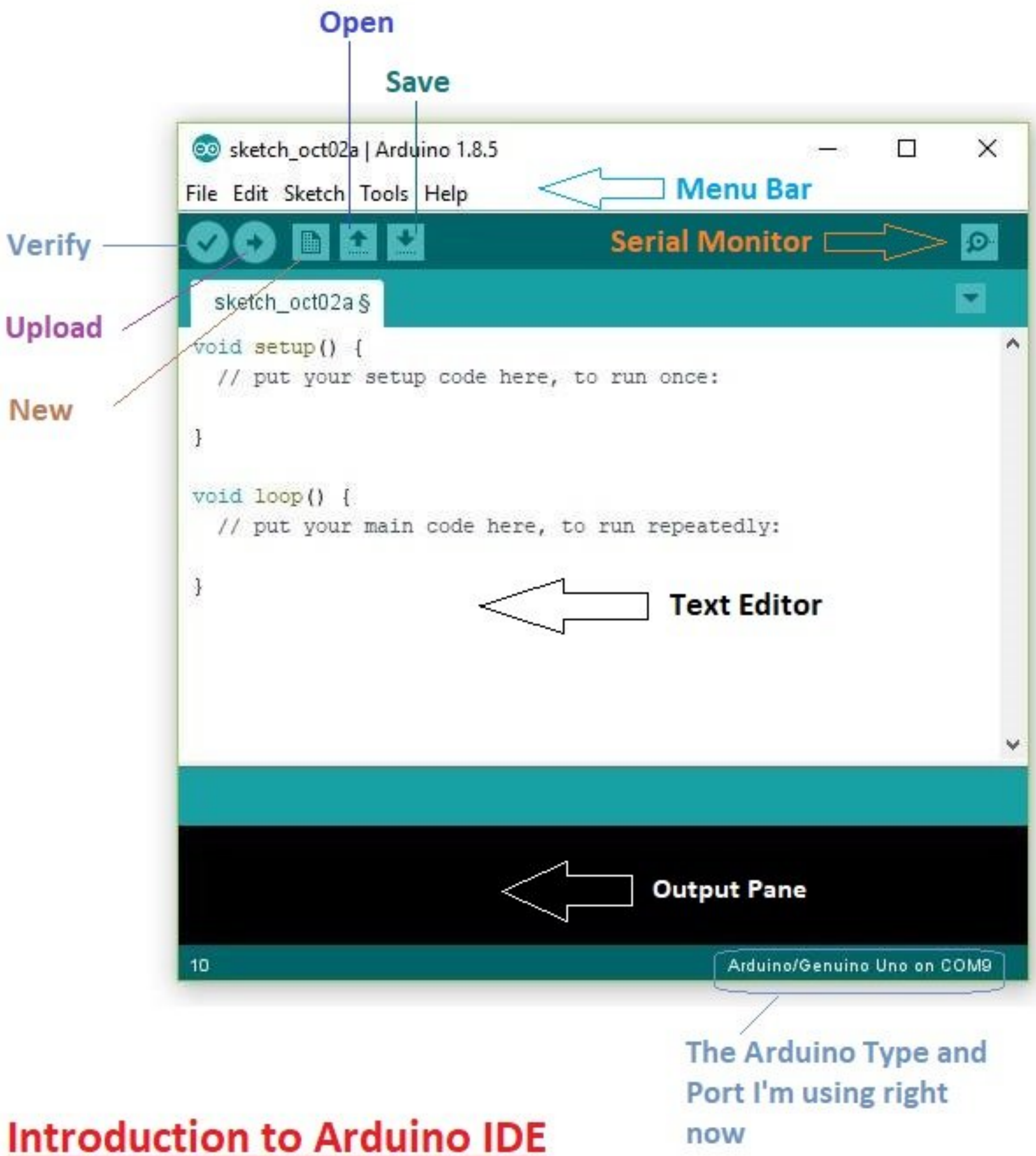
RESET
AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
DIGITAL (PWM~)
L
TX
RX
ARDUINO
UNO
ON
ICSP
WWW.ARDUINO.CC – MADE IN ITALY
POWER
ANALOG IN
IOREF
RESET
3.3V
5V
GND
GND
Vin
A0
A1
A2
A3
A4
A5



JustDoElectronics
DC Power Jack
USB Port
Reset Button
No Connection
5 V
Reset Input
3.3 V
5 V
Ground
Ground
Vin 7-12 V
Analog Pin 0 A0
Analog Pin 1 A1
Analog Pin 2 A2
Analog Pin 3 A3
I2C/SDA Analog Pin 4 A4
I2C/SCL Analog Pin 5 A5
I2C/SCL Serial Clock
I2C/SDA Serial Data
Analog Reference Voltage
Ground
13 Digital Pin13 SPI/SCK
12 Digital Pin12 SPI/MISO
11 Digital Pin11 SPI/MOSI PWM
10 Digital Pin10 SPI/SS PWM
9 Digital Pin9 PWM
8 Digital Pin8
7 Digital Pin7
6 Digital Pin6 PWM
5 Digital Pin5 PWM
4 Digital Pin4
3 Digital Pin3 Ext Int 1 PWM
2 Digital Pin2 Ext Int 0
1 Digital Pin1 Serial Port TXD
0 Digital Pin0 Serial Port RXD
www.prateeks.in


LED - Power-On
LED - TX &RX
LED - Load & Pin 13
Reset Button
Serial Port (USART)
14x Digital IN/OUT
(6x PWM OUT~)
(5V, 20mA continuous,
40mA max)
USB
(5V Power IN &
Serial Port for
Loading program)
Atmel ATmega328P
Microcontroller
(8-bit, 16 MHz,
32 KB Flash,
1 KB EEPROM,
2 KB SRAM)
DC Power Jack
(AC-to-DC adapter)
(7-12V)
Power OUT
(5V, <500mA)
(3.3V, 50mA)
5V Power IN
Power IN
(9V battery)
6x Analog IN
(0-5V 10-bit ADC)
or Digital IN/OUT
DIGITAL (PWM~)
UNO
ARDUINO
POWER
ANALOG IN

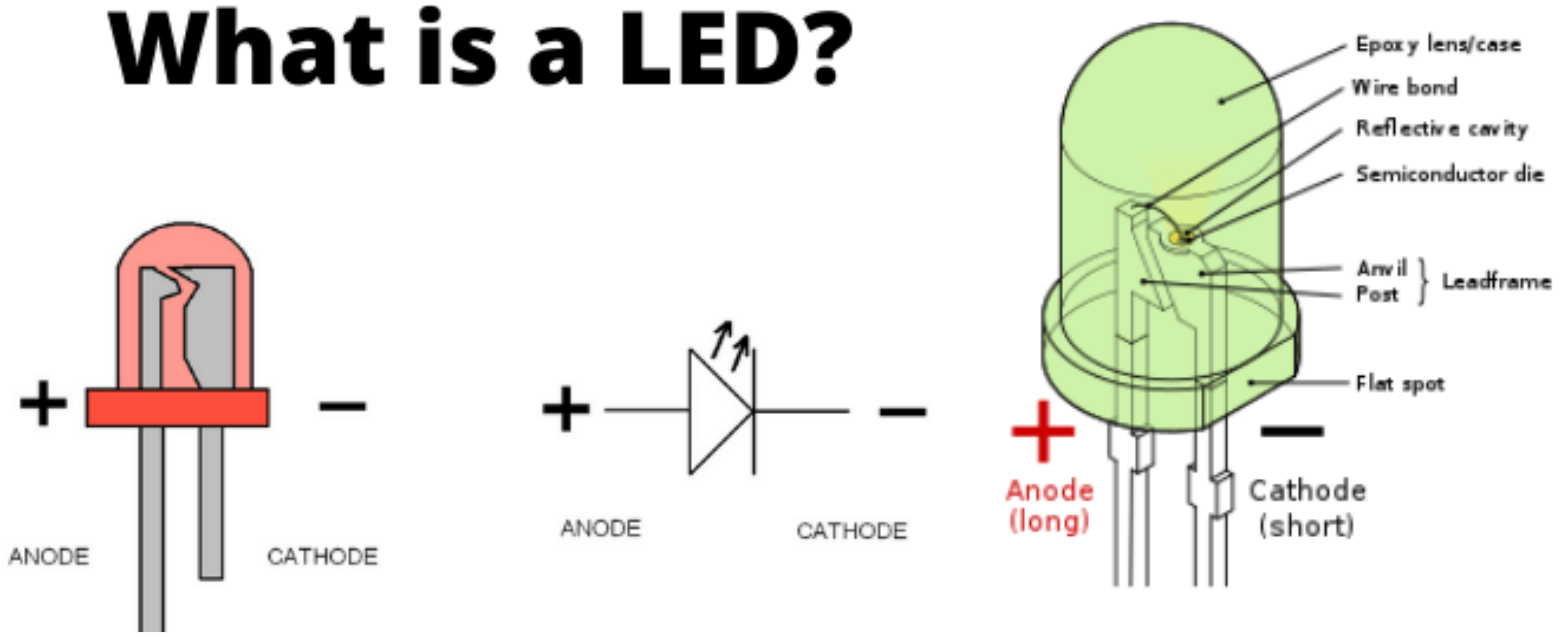
Arduino IDE

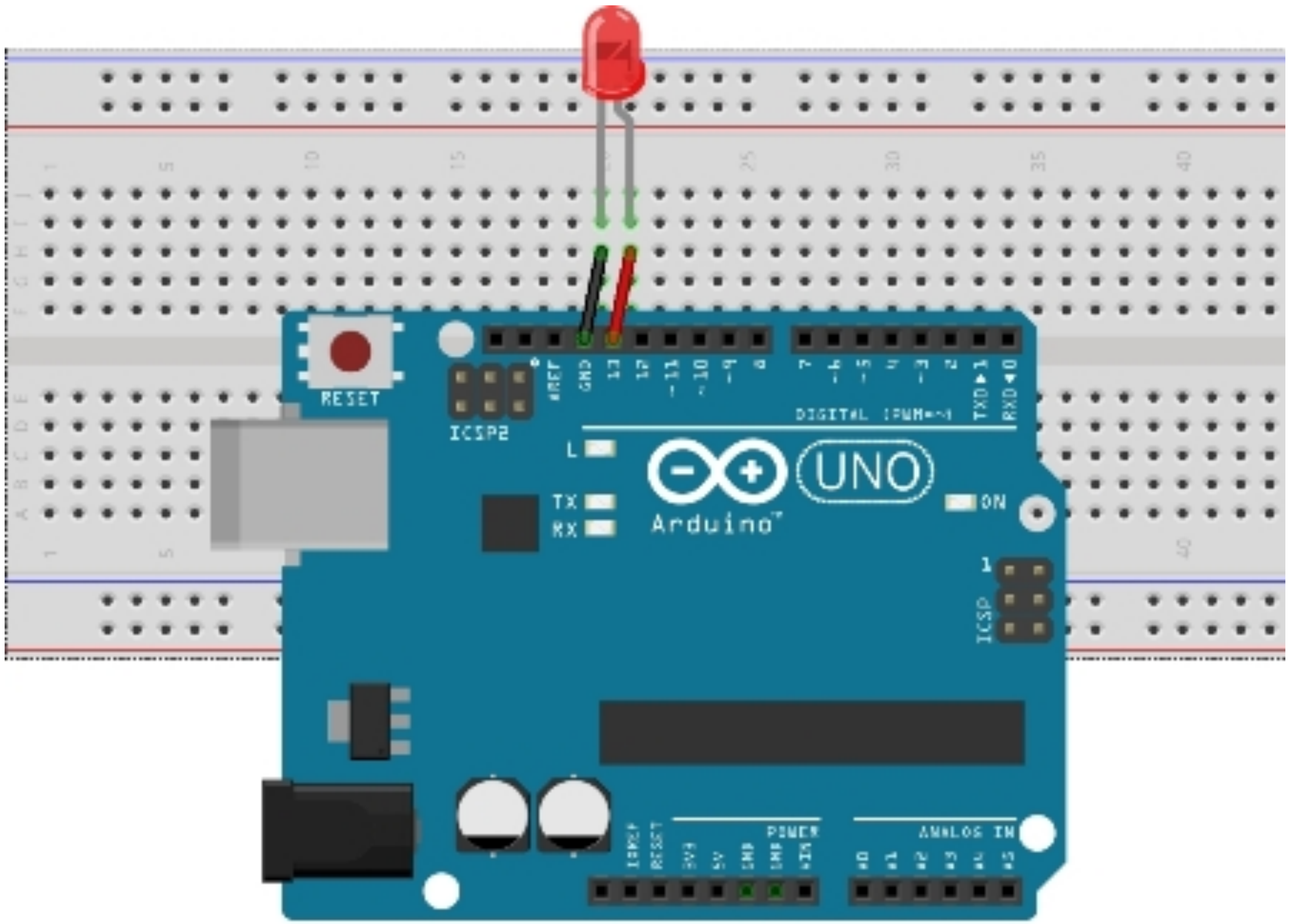


Introduction to Arduino IDE

Led Interfacing With Arduino

What is a LED?

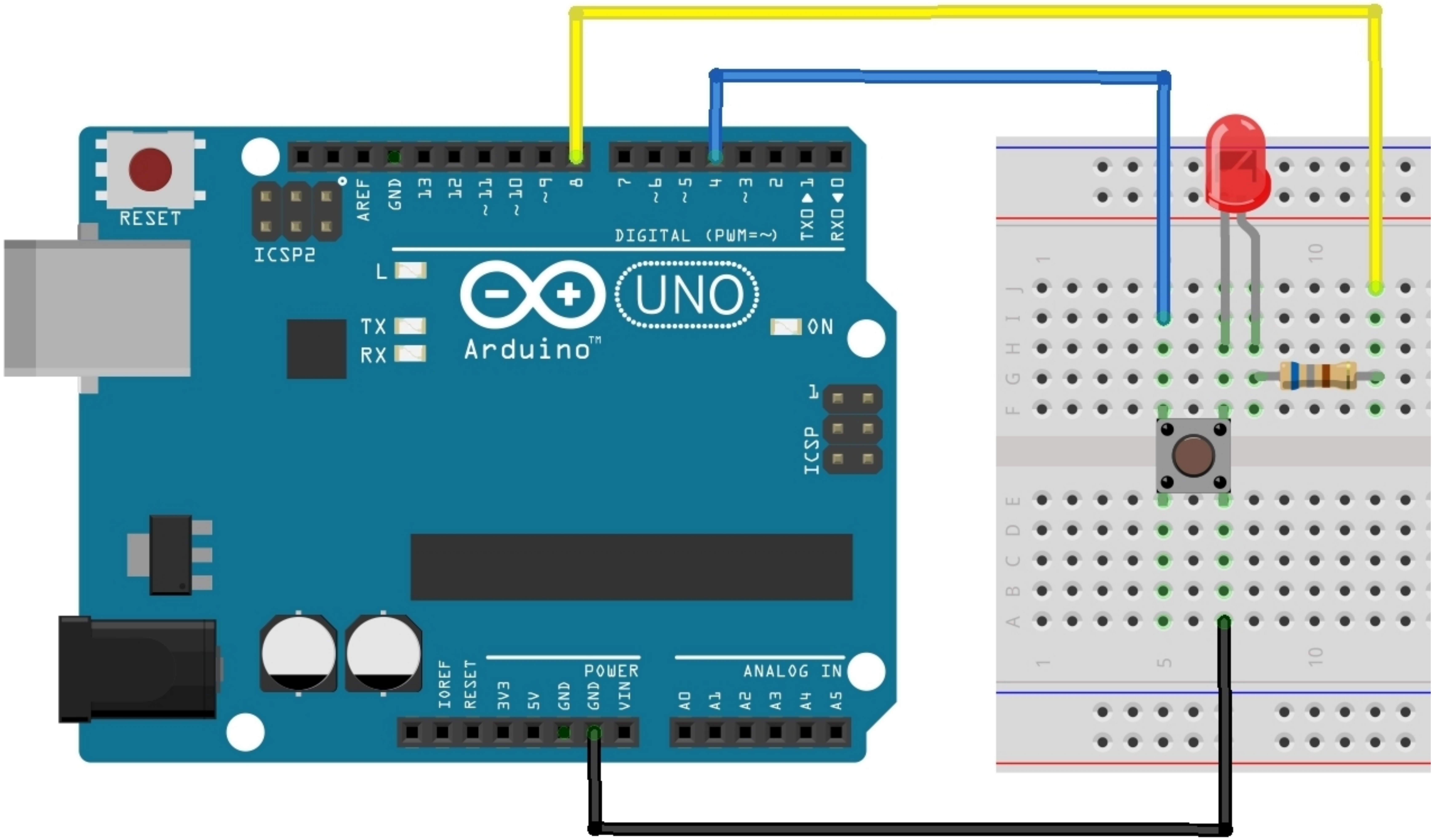




Code

```
//Prateek
//www.justdoelectronics.com

void setup() {
  pinMode(D4, OUTPUT);
}

void loop() {
  digitalWrite(D4, HIGH);
  delay(1000);
  digitalWrite(D4, LOW);
  delay(1000);
}
```

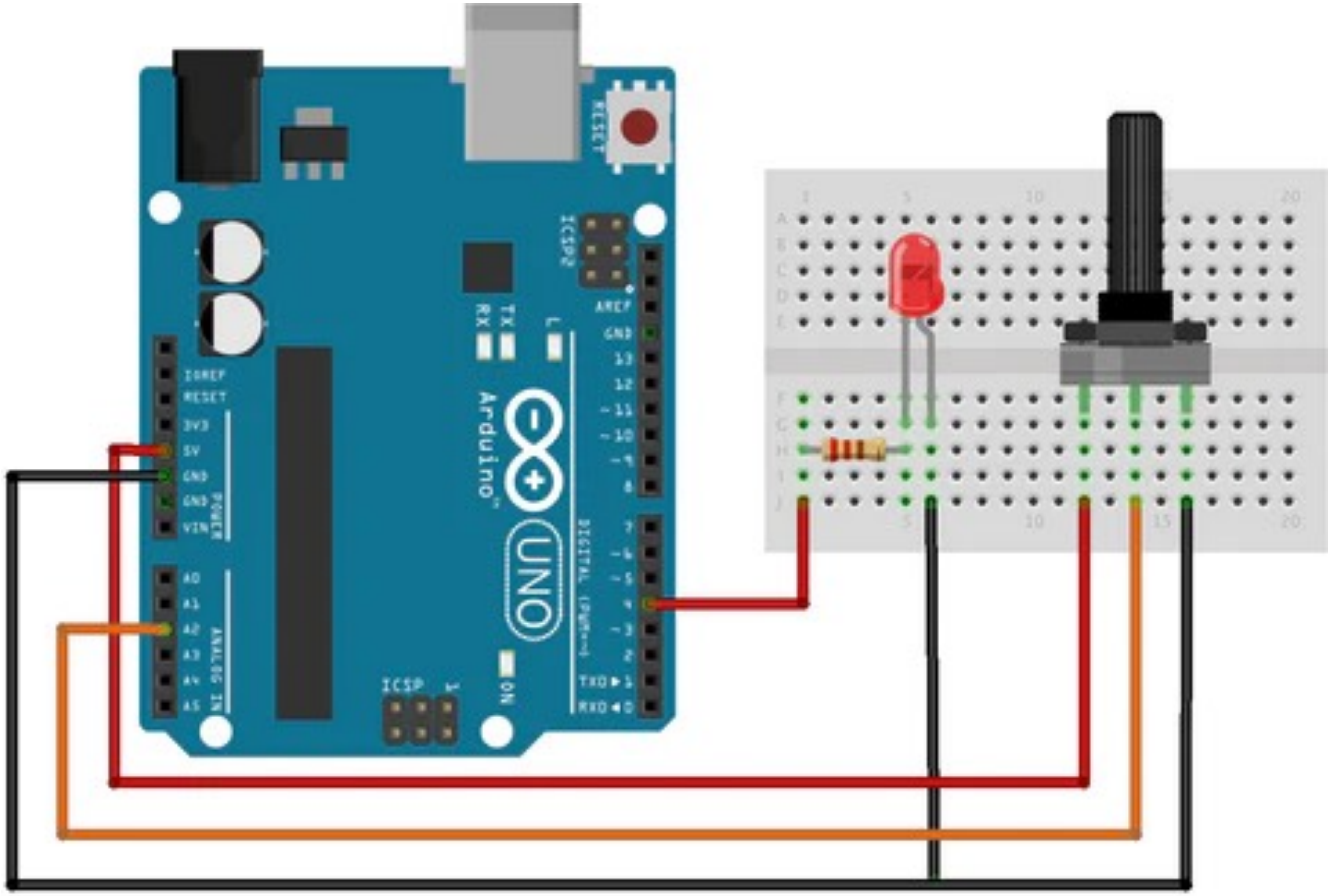
Push Button And Led Interfacing With Arduino



Code

```
const int switchPin = 4; //Switch Connected to PIN 4
const int ledPin = 8; //LED Connected to PIN 8
int switchState = 0; //Variable for reading switch status
void setup()
{
    pinMode(ledPin, OUTPUT); //LED PIN is Output
    pinMode(switchPin, INPUT_PULLUP); //Switch PIN is Input
}
void loop()
{
    switchState = digitalRead(switchPin); //Read the status of the Switch
    if (switchState == LOW) //If the switch is pressed
    {
        digitalWrite(ledPin, HIGH); //LED ON
        delay(3000); //3 Second Delay
        digitalWrite(ledPin, LOW); //LED OFF
    }
}
```

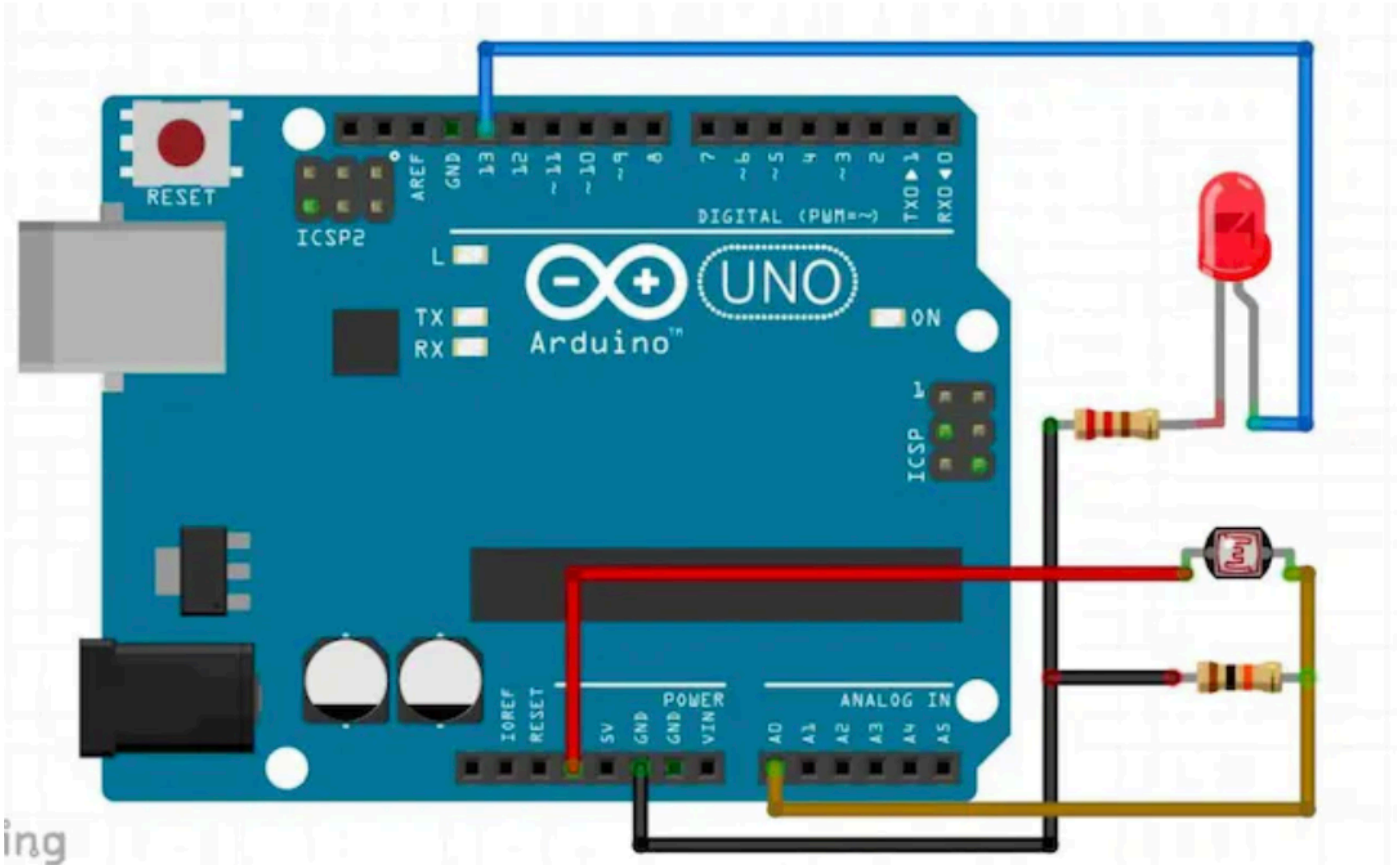
10k Pot Interfacing With Arduino UNO



Code

```
int potPin = A2;
int ledPin = 8;
int val=0;

void setup() {
pinMode (ledPin, OUTPUT);//Sets the LED pin as an output
}

void loop() {
val = analogRead(potPin); //reads the analogue voltage from the pot via ADC and stores it in the variable

digitalWrite(ledPin, HIGH); //turn the LED output pin high turning the LED on

delay(val); //waits an amount of time dependent on the potentiometer position

digitalWrite(ledPin, LOW); //turn the LED output pin low turning the LED off

delay(val); //waits an amount of time dependent on the potentiometer position

}
```

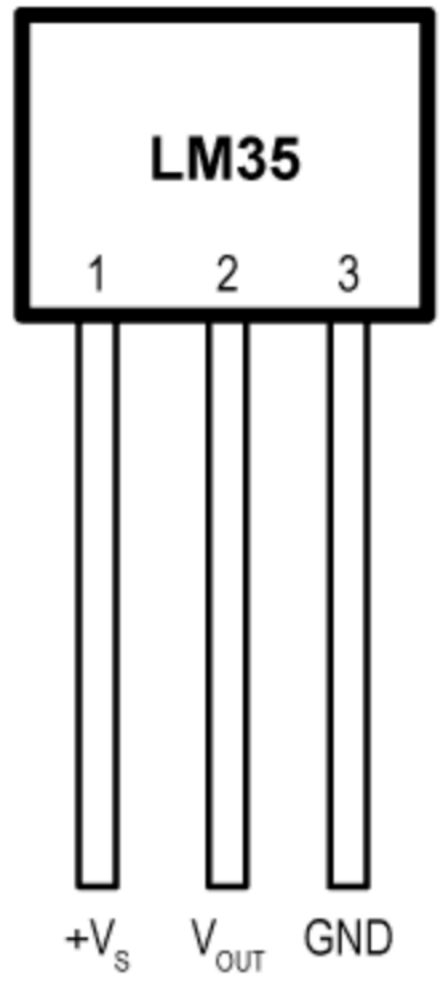
LDR With Arduino UNO

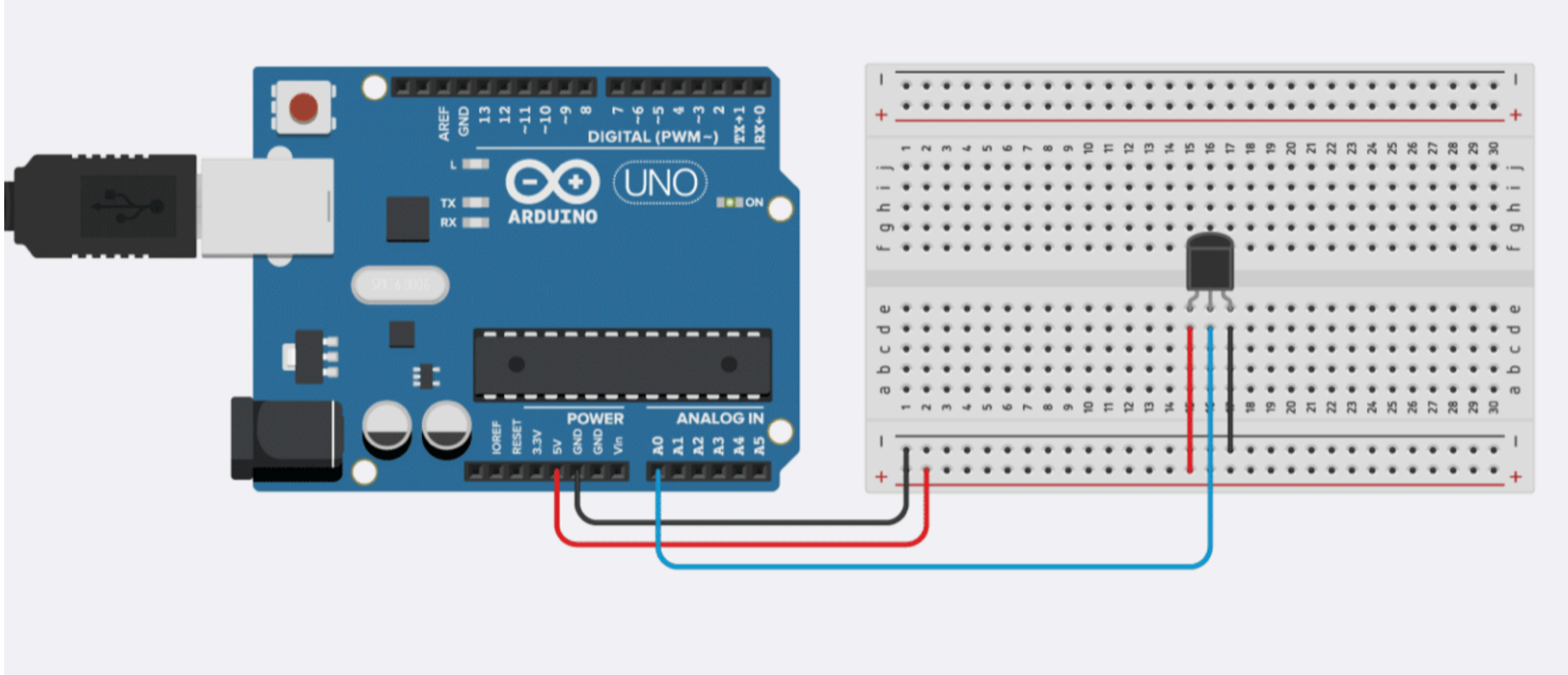


Code

```
const int ledPin = 13;

const int ldrPin = A0;

int threshold = 600;

void setup() {

  Serial.begin(9600);

  pinMode(ledPin, OUTPUT); //Make LED pin as output

  pinMode(ldrPin, INPUT); //Make LDR pin as input

}

void loop()
{

  int ldrStatus = analogRead(ldrPin);
  if (ldrStatus <= threshold)
  {

    digitalWrite(ledPin, HIGH);  //Turing LED ON

    Serial.print("Its DARK, Turn on the LED : ");

    Serial.println(ldrStatus);

  }

  else

  {

    digitalWrite(ledPin, LOW); //Turing OFF the LED

    Serial.print("Its BRIGHT, Turn off the LED : ");

    Serial.println(ldrStatus);

  }

}
```

LM35 With Arduino UNO

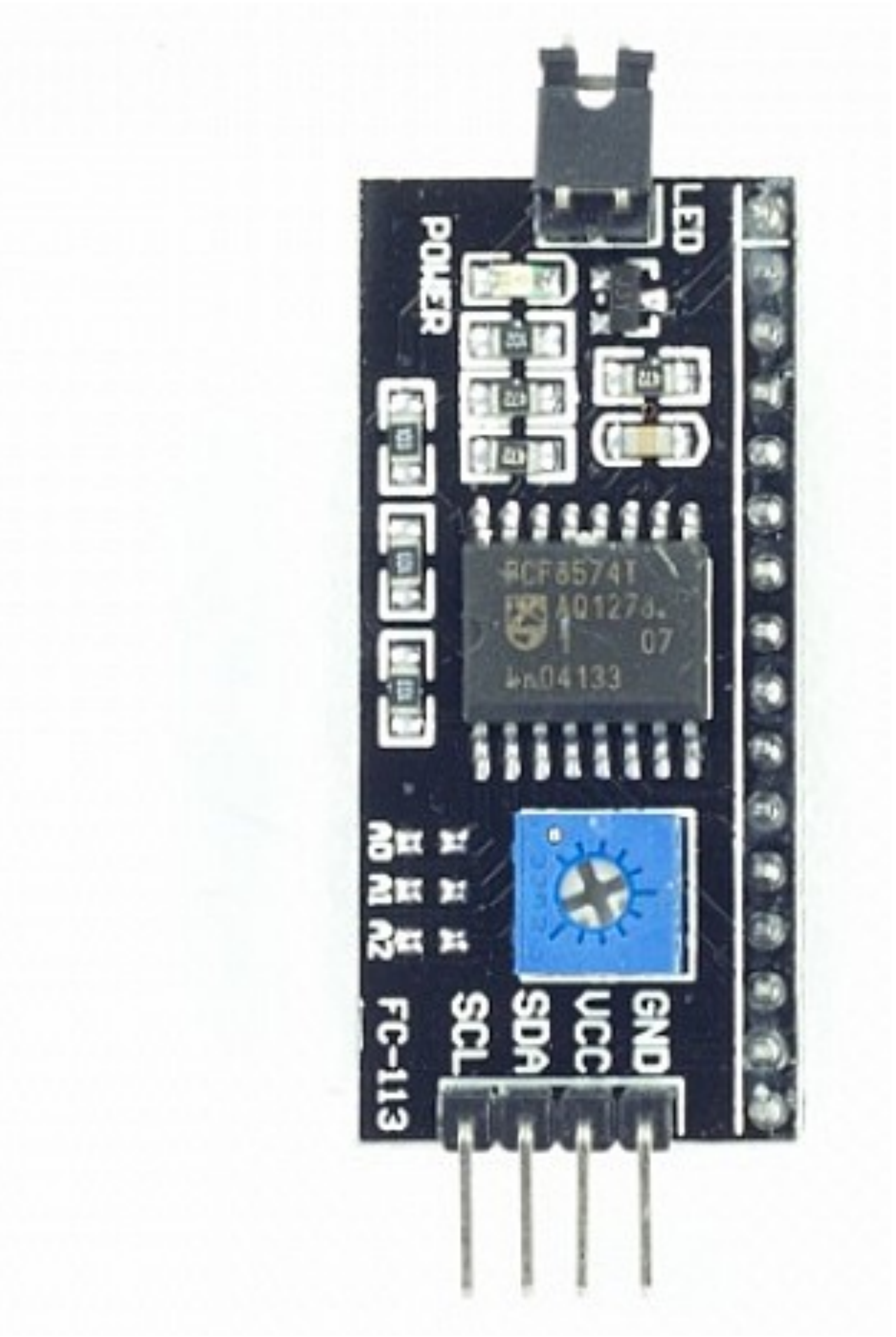


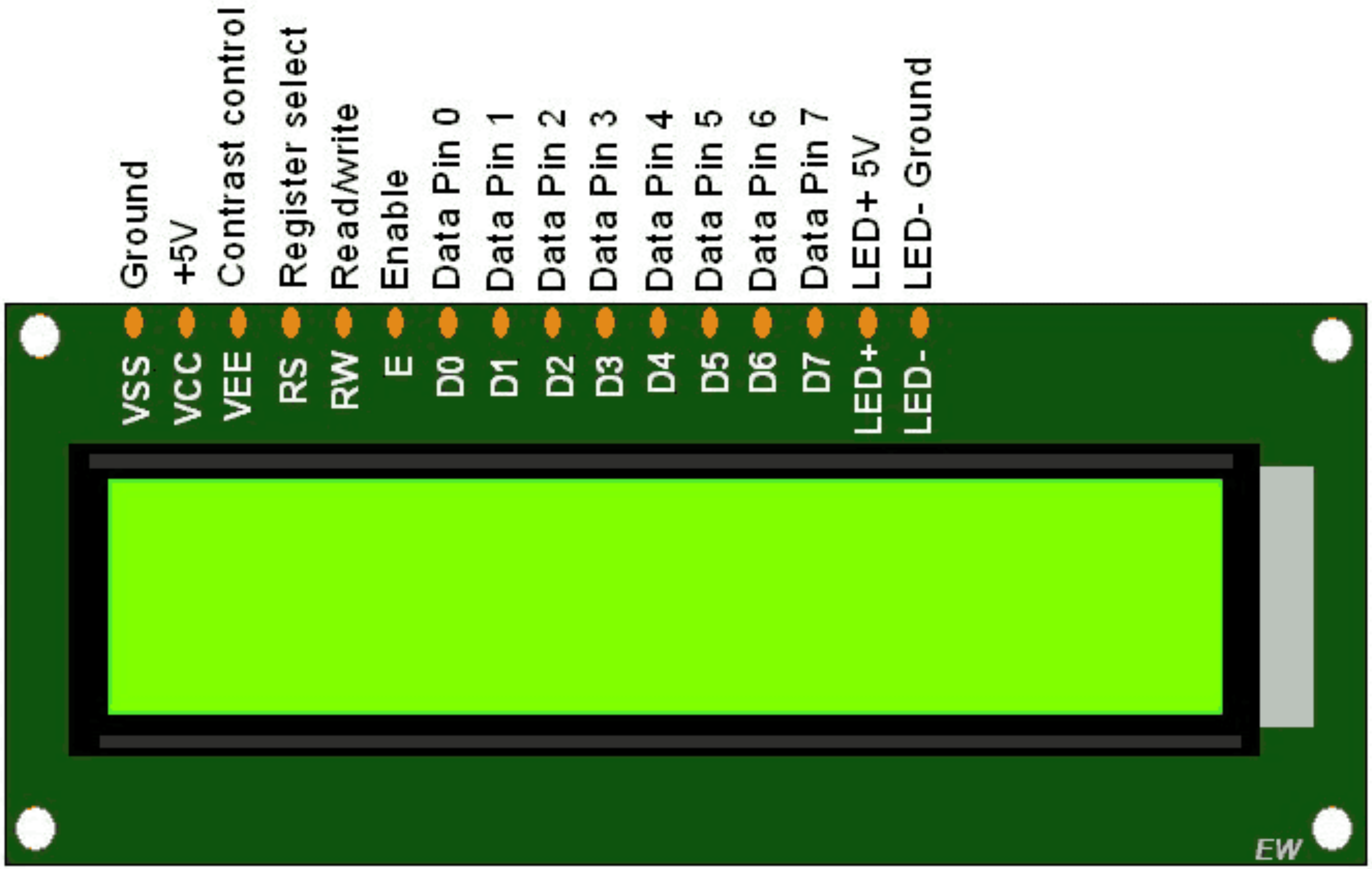


Code

```
#define sensorPin A0

void setup() {

  Serial.begin(9600);
}

void loop() {

  int reading = analogRead(sensorPin);
  float voltage = reading * (5000 / 1024.0);
  float temperature = voltage / 10;
  Serial.print(temperature);
  Serial.print(" \xC2\xB0");
  Serial.println("C");

  delay(1000);
}
```

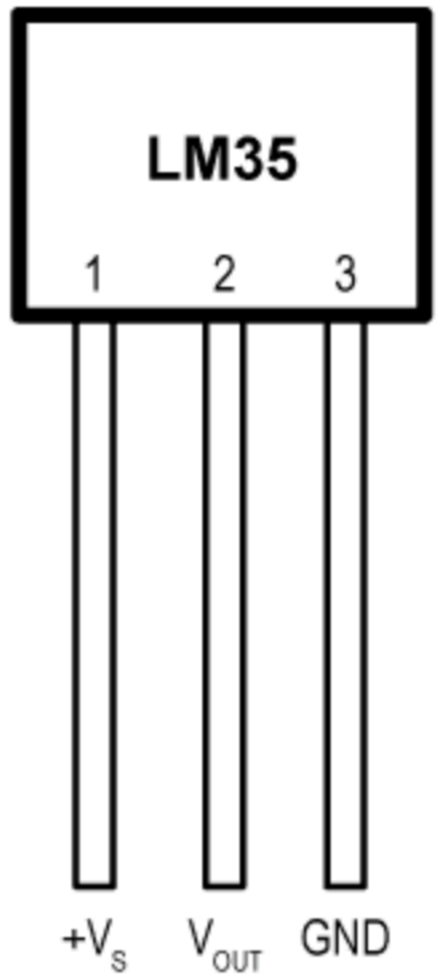
16x2 LCD Display

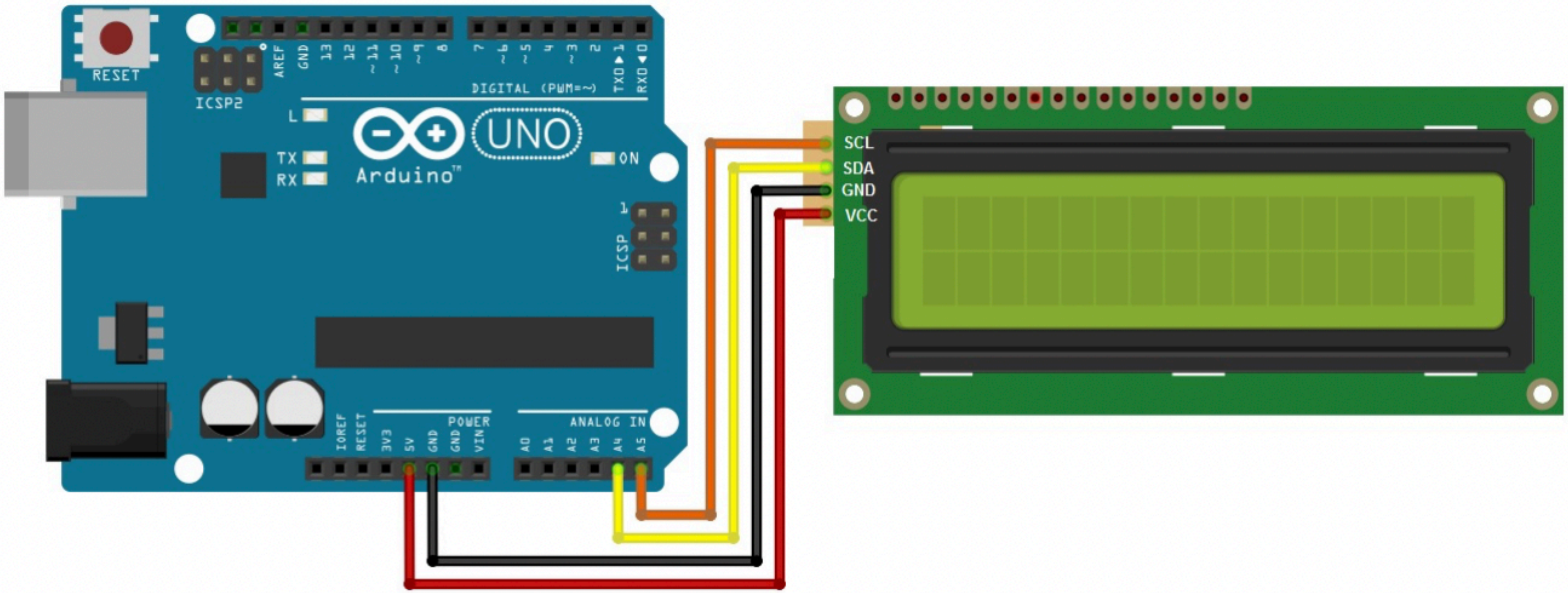




16x2 LCD Display

16x2 LCD With Arduino UNO

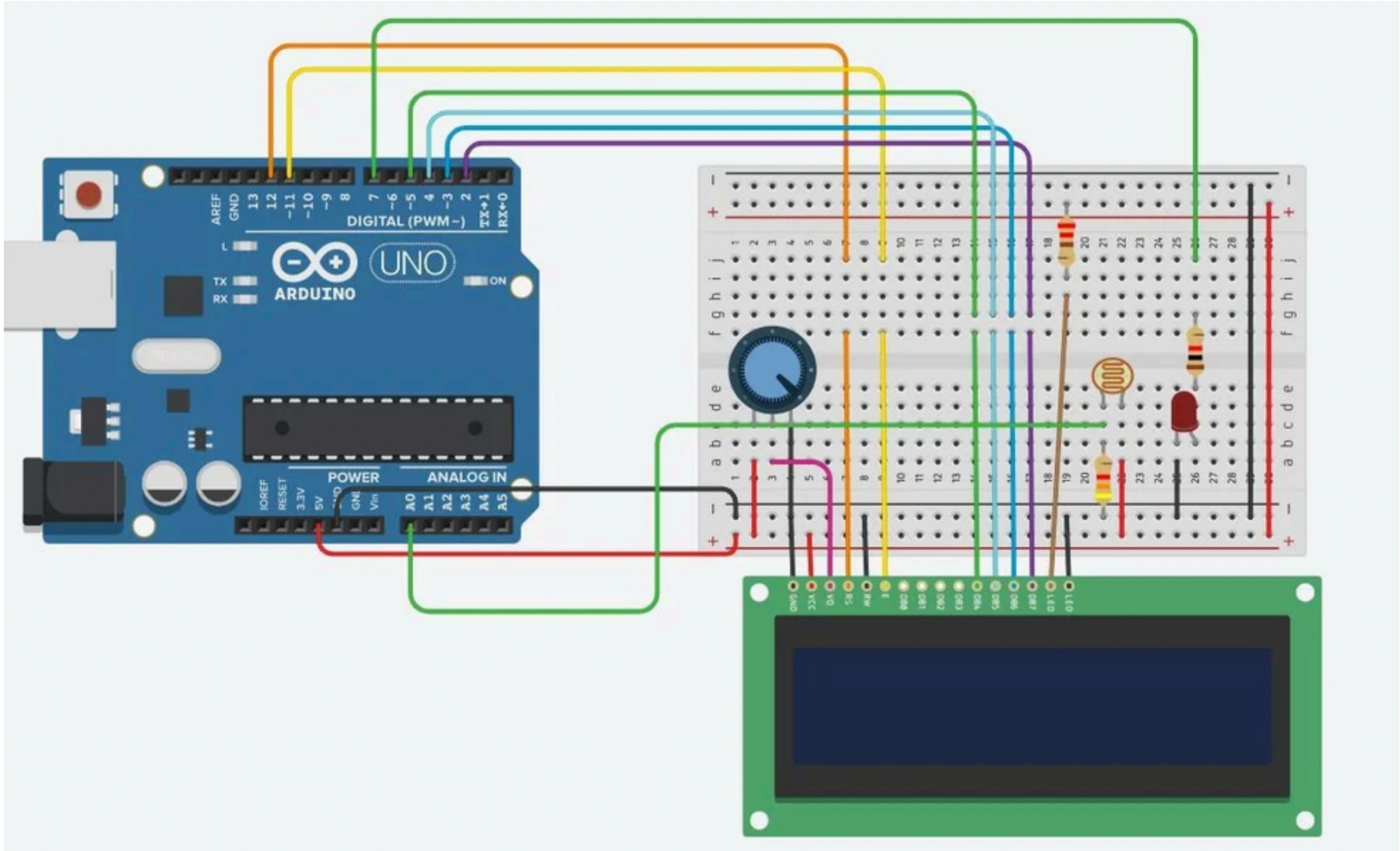




Code

```
//Prateek
//wwww.prateeks.in
//https://www.youtube.com/c/JustDoElectronics/videos

#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27, 16, 2);
void setup()
{
  Serial.begin(9600);
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("  Welcome To");
  lcd.setCursor(0, 1);
  lcd.print("JustDoElectronic");
  lcd.clear();

}
void loop()
{

}
```

16x2 LCD With LDR & Arduino UNO

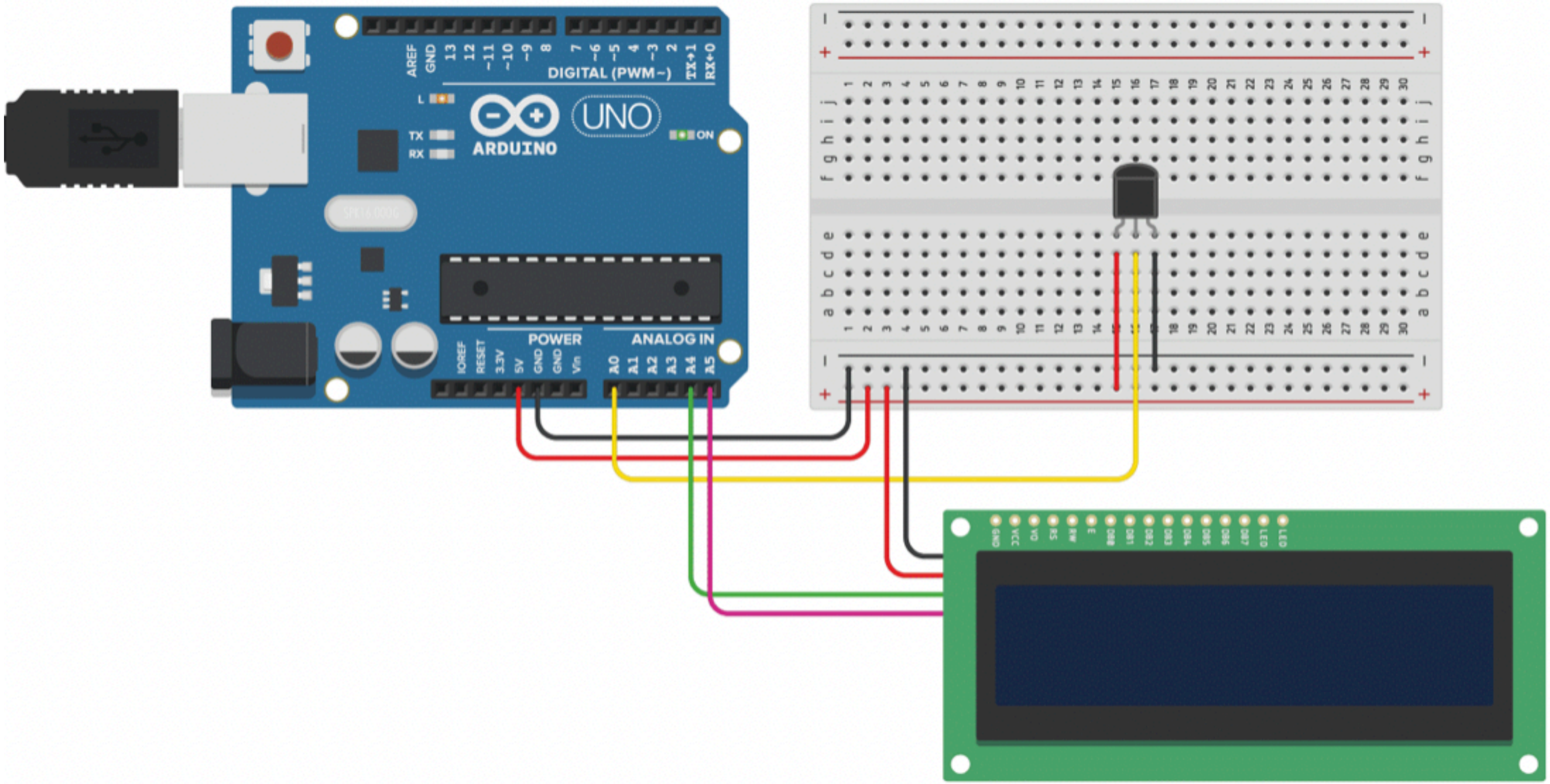


Code

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(12,11,5,4,3,2);
int sensorValue = 0;

int led = 7;

void setup()

{

pinMode(led,OUTPUT);

pinMode(A0,INPUT);

Serial.begin(9600);

lcd.begin(16,2);

lcd.setCursor(0,0);

lcd.print("SmartLightSystem");

}
```

```
void loop()

{

lcd.setCursor(0,0);

lcd.print("SmartLightSystem");

sensorValue= analogRead(A0);

Serial.println(sensorValue);

delay(100);

if (sensorValue < 750)

{

lcd.setCursor(0,1);

lcd.print("LED ON");

digitalWrite(led,HIGH);

delay(100);

}

else

{

lcd.setCursor(0,1);

lcd.print("LEDOFF");

digitalWrite(led,LOW);

delay(100);

}

}
```

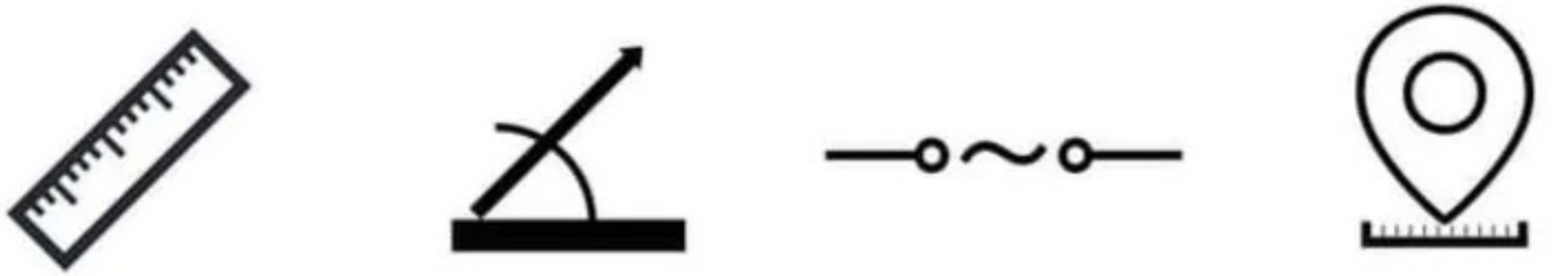
16x2 LCD With LM35 & Arduino UNO



Code

```
#include <LiquidCrystal_I2C.h> //If you use I2c adapter.
#define LM35 A2
#define RED 7
#define GREEN 6
byte degree[8] =
{
0b00011,
0b00011,
0b00000,
0b00000,
0b00000,
0b00000,
0b00000,
0b00000
};

float lm_value;
float tempc;
LiquidCrystal_I2C lcd (0x3F, 16,2);
void setup() {
  Serial.begin(9600);
  lcd. init ();
  lcd. backlight ();
  pinMode(RED, OUTPUT);
  pinMode(GREEN, OUTPUT);
lcd.createChar(1, degree);
lcd.setCursor(0,0);
lcd.print(" Digital ");
lcd.setCursor(0,1);
lcd.print(" Thermometer ");
delay(2000);
lcd.clear();
}
void loop() {
  lm_value = analogRead(LM35);
  tempc = (lm_value * 500) / 1023;
  Serial.println(tempc);
lcd.setCursor(2,0);
lcd.print("Temperature");
lcd.setCursor(4,1);
lcd.print(tempc);
lcd.write(1);
lcd.print("C");
  if (tempc > 60) {
    digitalWrite(RED, HIGH);
    digitalWrite(GREEN, LOW);
  }
  else {
    digitalWrite(GREEN, HIGH);
    digitalWrite(RED, LOW);
  }
  delay(1000);
}
```

Ultrasonic Sensor (HCSR-04)

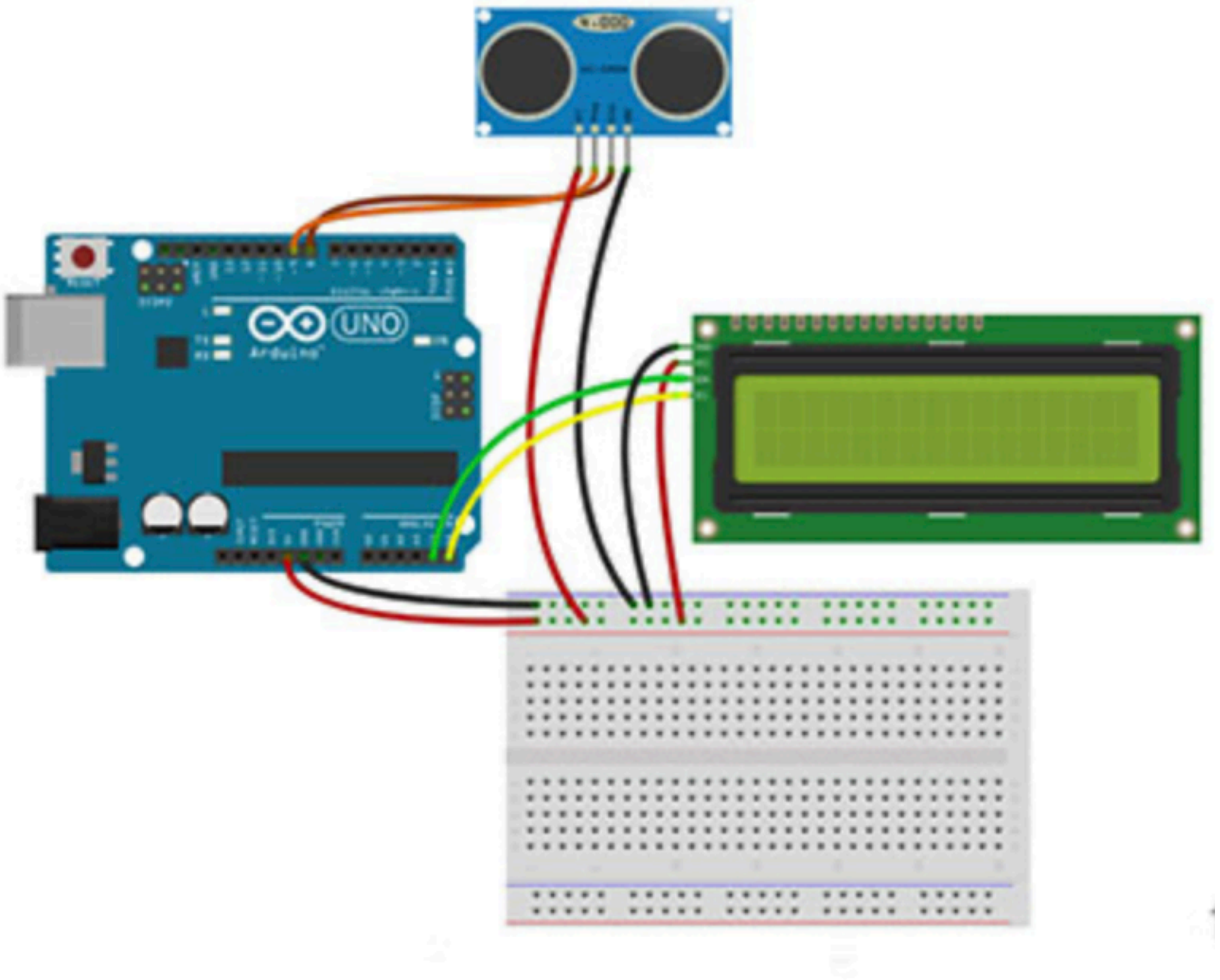


0.3 CM	<15'	<2MA	2-450CM
RESOLUTION	ANGLE	CURRENT	DETECTION RANGE

1 2 3 4

1. VCC
2. TRIG
3. ECHO
4. GND

16x2 LCD With Ultrasonic Sensor & Arduino UNO

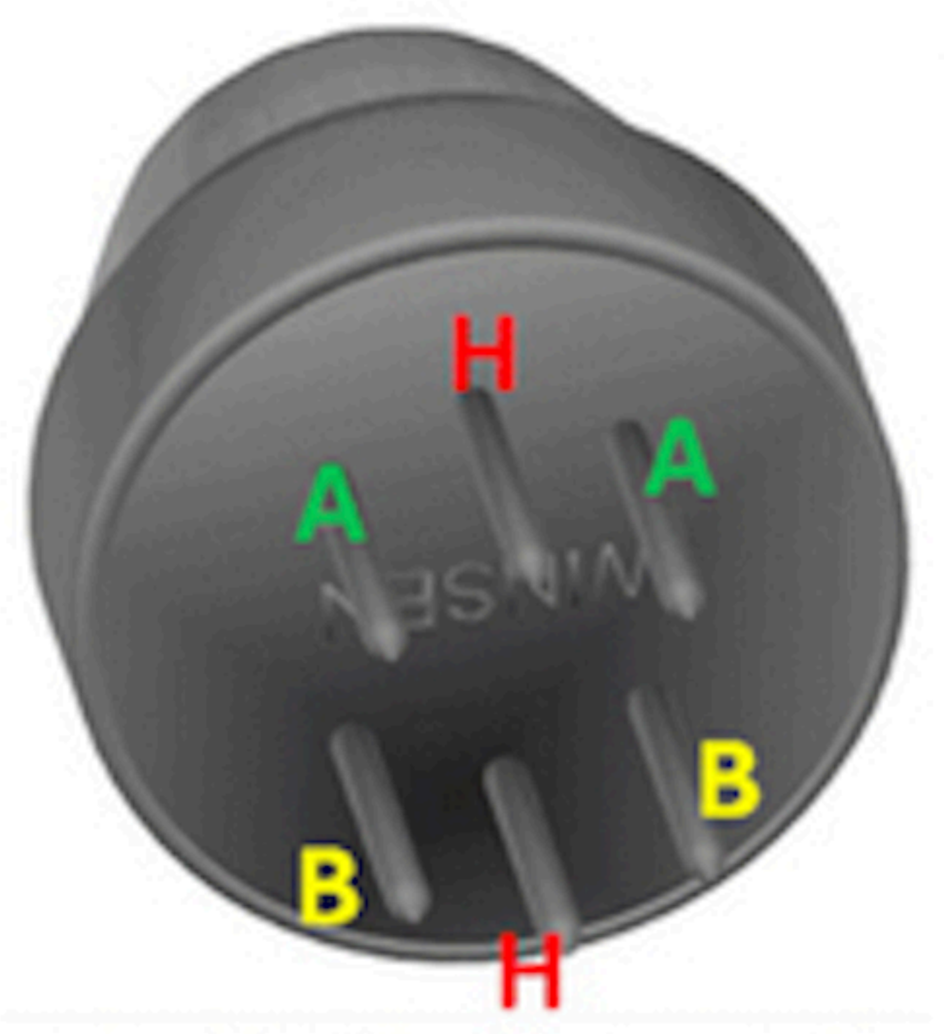


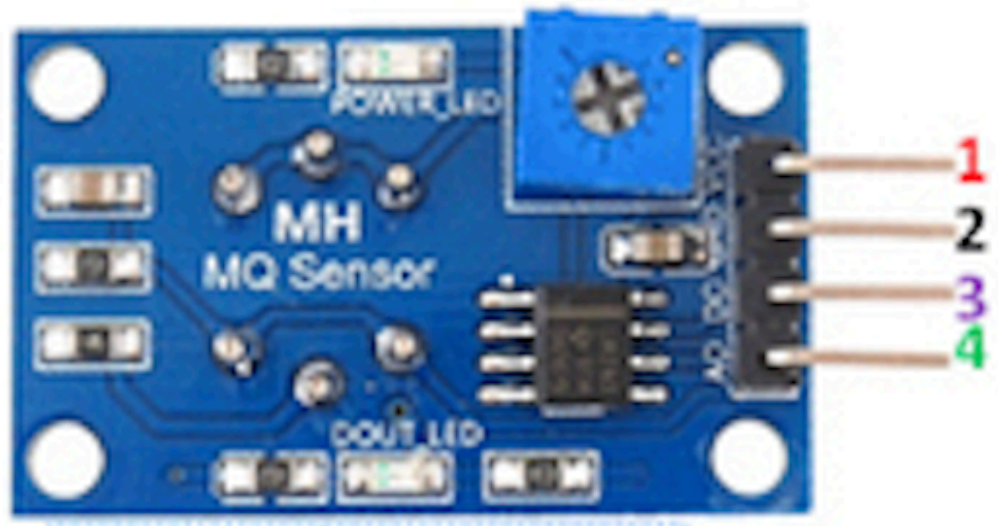
Code

```
#include <Wire.h>
#include "LiquidCrystal_I2C.h"

#define trig 9
#define echo 8

LiquidCrystal_I2C lcd(0x27, 16, 2);
float duration,dist_inch;
int dist_cm;

void setup() {

Serial.begin(9600);
lcd.clear();

lcd.backlight();
lcd.begin();

lcd.setCursor(0, 0);
lcd.print("dist. cm :");
lcd.setCursor(0, 1);
lcd.print("dist. inch:"); /
}
```

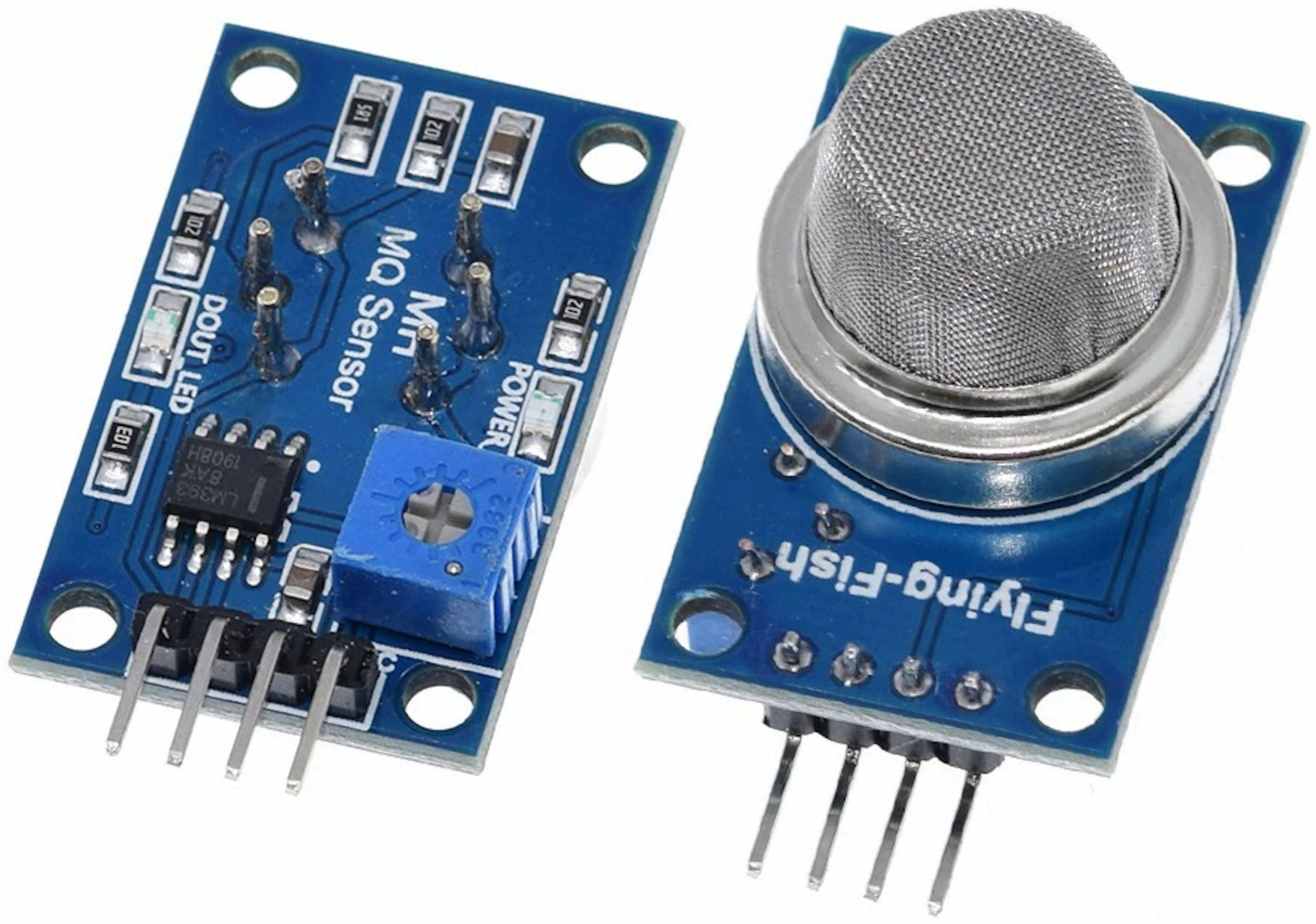
```
void loop() {

digitalWrite(trig, LOW);
delayMicroseconds(2);
digitalWrite(trig, HIGH);
delayMicroseconds(10);
digitalWrite(trig, LOW);
duration = pulseIn(echo, HIGH);
dist_cm = duration/58;
dist_inch = dist_cm/2.54;
lcd.setCursor(11, 0);
lcd.print(dist_cm);
lcd.setCursor(11, 1);
lcd.print(dist_inch);
delay(500);

// Clear the LCD
lcd.setCursor(11, 0);
lcd.print(" ");
lcd.setCursor(11, 1);
lcd.print(" ");
delay(10);
}
```

GAS Sensor

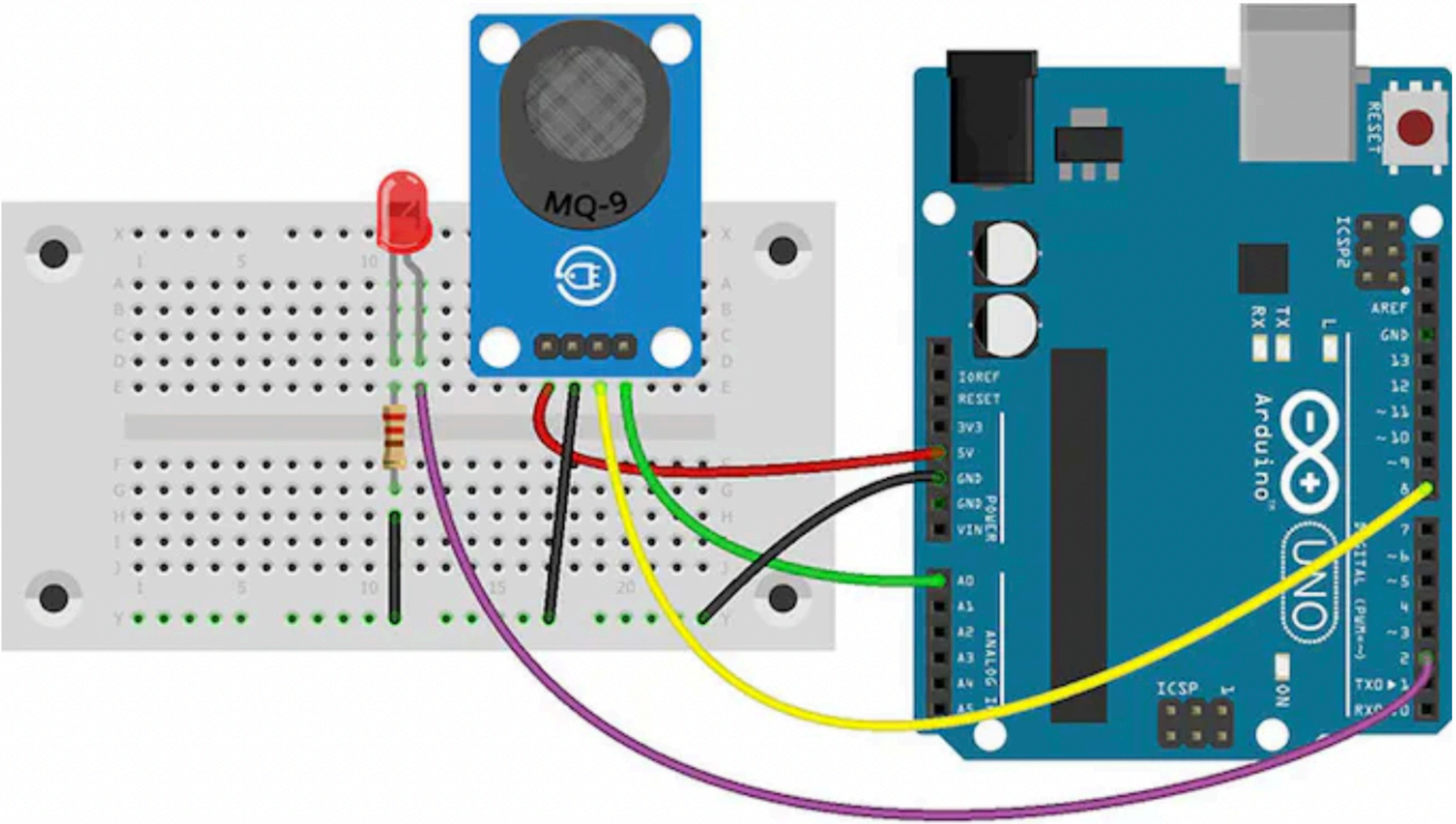




Pin No.	Pin Name
1	Vcc(+5V)
2	Ground
3	Digital Out
4	Analog out



16x2 LCD With GAS Sensor & Arduino UNO



Code

```
const int LED = 2;
const int DO = 8;
void setup() {
 Serial.begin(9600);
 pinMode(LED, OUTPUT);
 pinMode(DO, INPUT);
}
void loop() {
 int alarm = 0;
 float sensor_volt;
 float RS_gas;
 float ratio;
 float R0 = 0.91;
 int sensorValue = analogRead(A0);
 sensor_volt = ((float)sensorValue / 1024) * 5.0;
RS_gas = (5.0 - sensor_volt) / sensor_volt;
 ratio = RS_gas / R0;
 Serial.print("sensor_volt = ");
 Serial.println(sensor_volt);
 Serial.print("RS_ratio = ");
 Serial.println(RS_gas);
 Serial.print("Rs/R0 = ");
 Serial.println(ratio);
 Serial.print("\n\n");
 alarm = digitalRead(DO);
 if (alarm == 1) digitalWrite(LED, HIGH);
 else if (alarm == 0) digitalWrite(LED, LOW);
 delay(1000);
}
```